

— Time Series I: EDA

Agenda

- What is a time series?
- Pandas tips & tricks
- Time series notations
- Backshifting and diffing
- Stationarity
- Autocorrelation and partial autocorrelation



Definition

A **time series** x_t is a set of observations being recorded at equally spaced intervals, each one at a specific time t .

Let's start out right away with some Pandas tips and tricks!



— Time Series Notations

Notation

Since a time series is a series, we sometimes denote it in mathematical series notation:

- $\{X_t\}$ as a random variable
- $\{x_t\}$ as realizations of $\{X_t\}$

When we model TS data later, the observations themselves are the response data. Still, you will often see the letter x used instead of y .

Notation

For example, if x_t is daily data, then, with respect to t :

$$x_{t-1}$$

is yesterday

$$x_t$$

is today

$$x_{t+1}$$

is tomorrow

Backshifting

The **backshift operator**, B , is an operation we will use often (for reasons to be revealed soon). It simply shifts our data “back” one time period:

$$BX_t = X_{t-1}$$

and can be applied multiple times:

$$B^2 X_t = BX_{t-1} = BX_{t-2}$$

In Pandas, the backshift operator is the method **`.shift()`**

Differencing

Another common operation is **differencing**: the act of taking row-to-row differences. You can think of this as “today’s change from yesterday”:

$$\nabla X_t = X_t - X_{t-1}$$

The Pandas method for this is **.diff()**.

Repeated Differencing

Note that differencing twice may not be what you think:

$$\begin{aligned}\nabla^2 X_t &= \nabla(X_t - X_{t-1}) \\ &= (X_t - X_{t-1}) - (X_{t-1} - X_{t-2}) \\ &= X_t - 2X_{t-1} + X_{t-2}\end{aligned}$$

Differencing will be important for achieving **stationarity**.

— Stationary Time Series

Stationarity

A time series is **stationary** if there is no overall trend in the data.

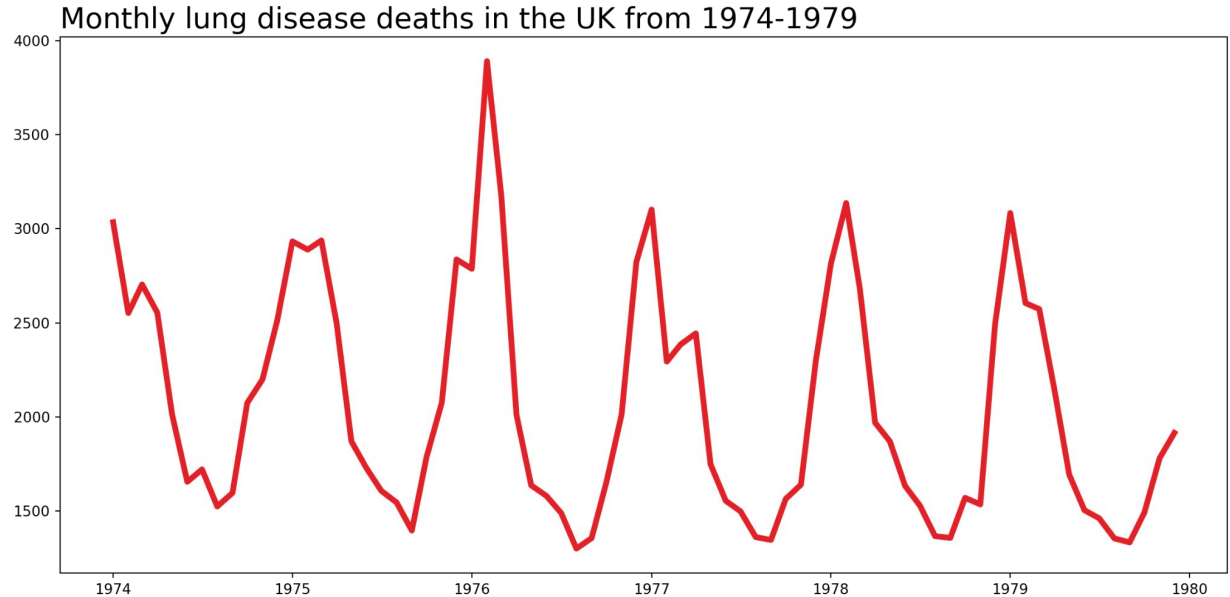
Stationarity will be a requirement of all time series modeling we do.

However, most time series data are not stationary.

Often times, **differencing** the data once will create a stationary time series. Sometimes, twice. You rarely need to difference more than twice.

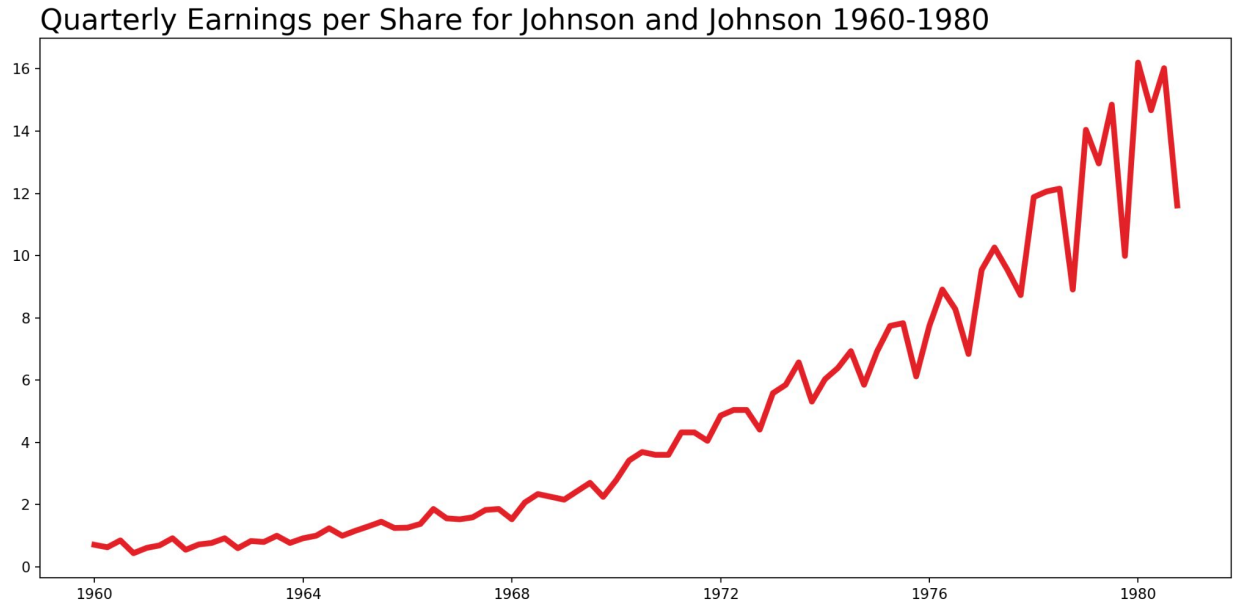
A stationary time series

This time series exhibits no overall upward or downward trend over time. There's a seasonal trend, but that doesn't prevent stationarity.



A nonstationary time series

This time series exhibits a clear upward trend over time and is thus **nonstationary**.

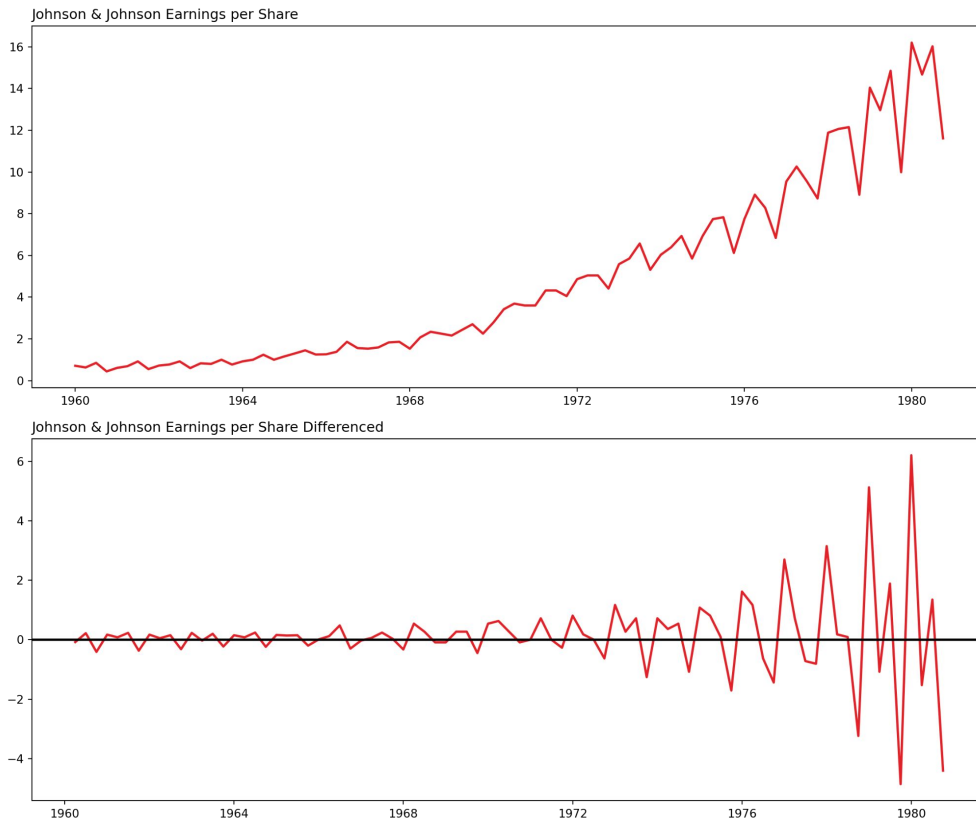


Differencing to (almost) achieve stationarity

After differencing the JJ data, we lose the overall upward trend.

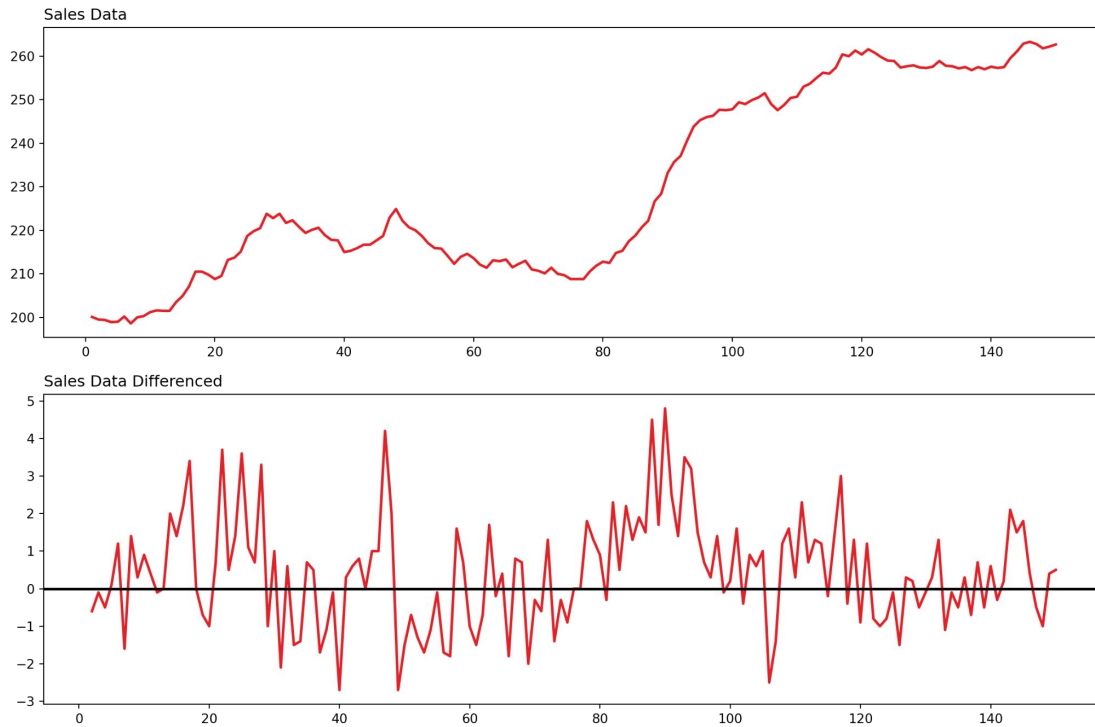
However, the variance increases over time, **so we still don't technically have stationarity.**

A solution here could be to log-transform our data before differencing.



Differencing to (almost) achieve stationarity

This is different sales data (Box-Jenkins, 1976). Here, differencing the data clearly results in a stationary time series with no trend or change in variance.



— Autocorrelation

Autocorrelation



Judging by the name, what do you think autocorrelation is?

Autocorrelation

Autocorrelation is a stationary time series' correlation with itself at a given lag h .

$$\rho(h) = \text{Corr}(X_t, X_{t-h})$$

Notice that $\rho(h)$ does not depend on t since we're assuming stationarity.

Sample Autocorrelation (in Pandas)

The **sample autocorrelation** at lag h can be computed in Python as:

```
np.corrcoef(x[1:], x.shift(h)[1:])
```

Autocorrelation in words

Simply put, the autocorrelation at lag 1 might be interpreted as *“How strong is the relationship of every day’s value versus its value yesterday?”*

At lag 7, maybe it’s *“How strong is the relationship between today and one week ago?”*

Examples

- Let x_t be the number of customers a restaurant sees on a daily basis. You're likely to see a 7-lag autocorrelation here, since restaurants see more customers on Fridays and Saturdays, and not so much on Mondays and Tuesdays.
- Let x_t be the monthly sales numbers at Toys R Us (RIP). You're likely to see a 12-lag autocorrelation since the December-to-December relationship is high.

Examples

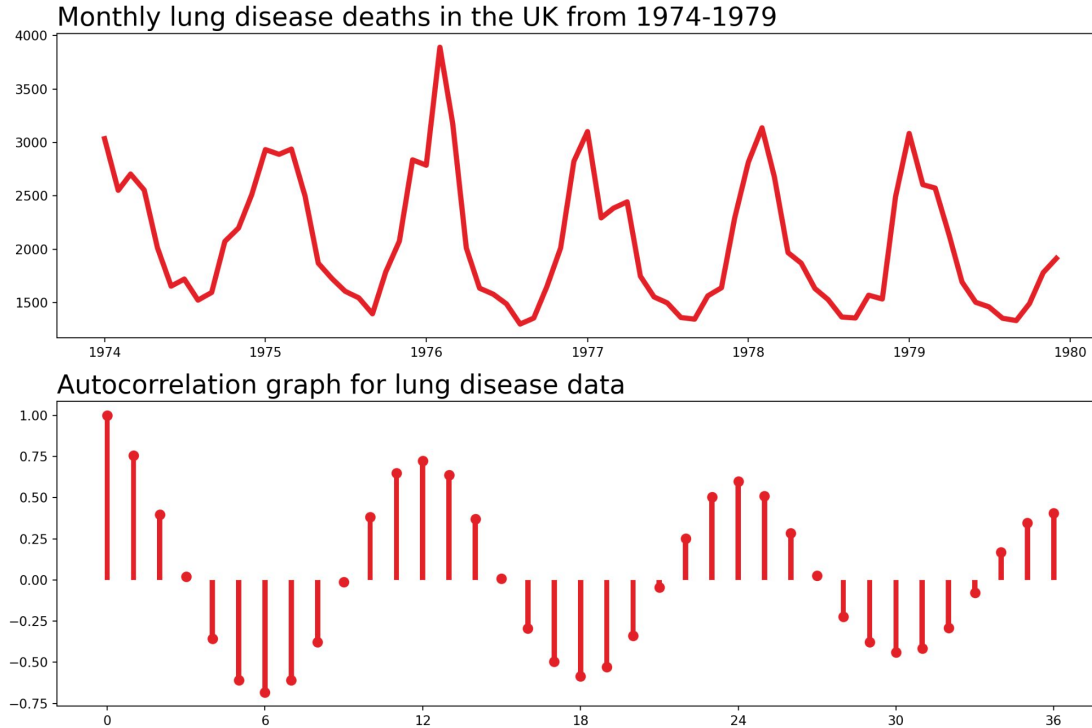
- Let x_t be the daily expenditure of a given person. You're likely to see 7-day autocorrelations (you spend more on weekends), 14-day autocorrelations (you spend more after you get paid bimonthly), and 365-day autocorrelations (you have yearly holidays in which you spend more).

Autocorrelation Graphs

An **autocorrelation graph** typically shows vertical line segments showing the size of the autocorrelation on the y -axis, and on the x -axis is the lag h .

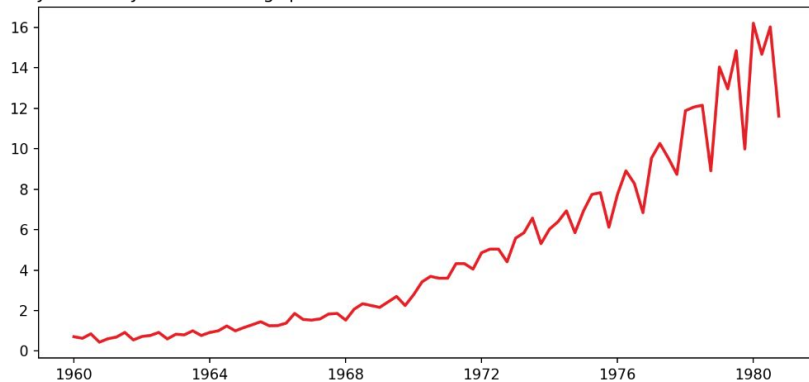
Often, they'll have a shaded region to show the boundaries of 95% confidence intervals for these autocorrelations.

Example ACF: Lung Disease (monthly data)

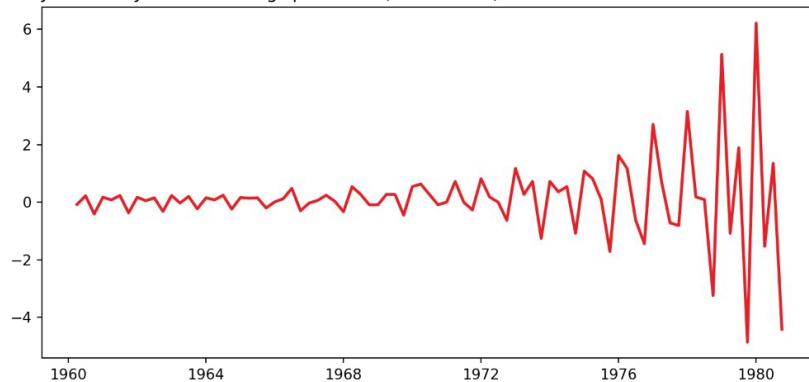


Example ACF: J&J Earnings (increasing trend)

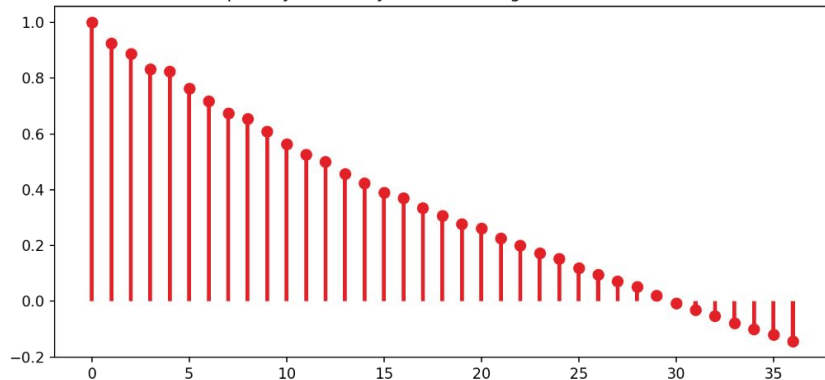
Johnson & Johnson Earnings per Share



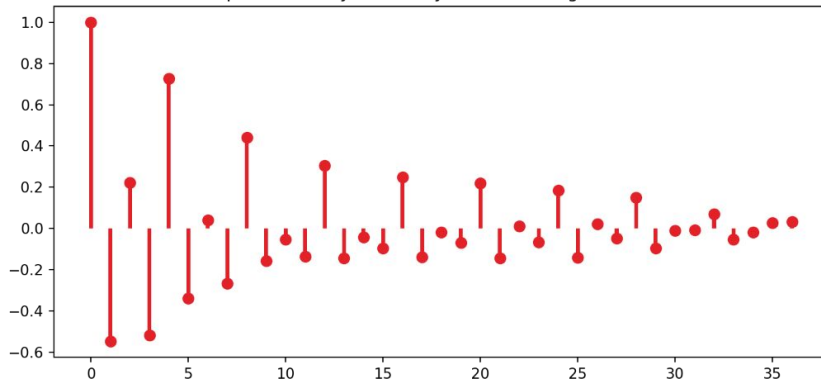
Johnson & Johnson Earnings per Share (Differenced)



Autocorrelation Graph for Johnson & Johnson Earnings



Autocorrelation Graph for Dified Johnson & Johnson Earnings



Partial Autocorrelation

The **partial autocorrelation function** is the same as the autocorrelation function, but it controls for all lower lags. The math is much more difficult (and much less intuitive) to write out, but an example can help:



Partial Autocorrelation

Let x_t be Tim's daily calorie consumption. He normally consumes about 2,500 calories per day, but on Sundays, he has a massive brunch with all of his friends. The brunch consists of all you can eat of eggs benedict, patatas bravas, corned beef hash, and of course bottomless mimosas, totalling about 5,000 calories.



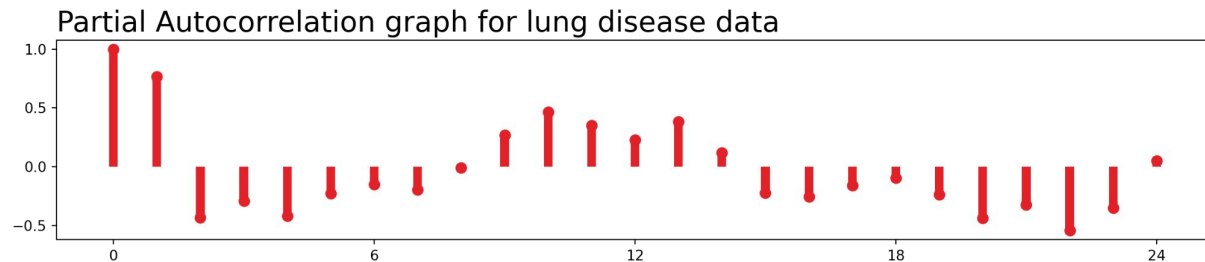
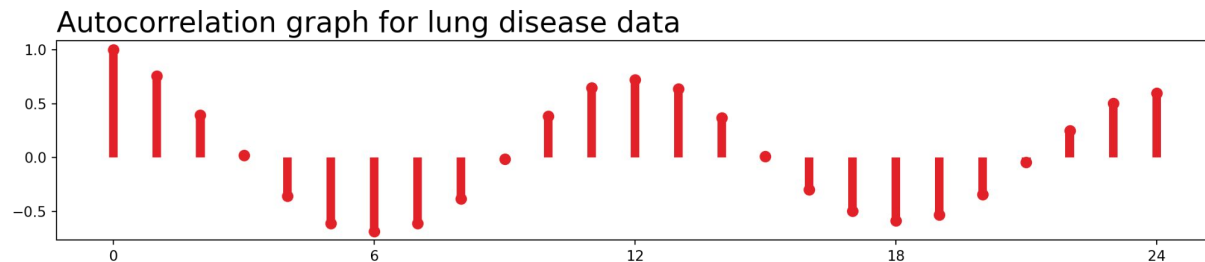
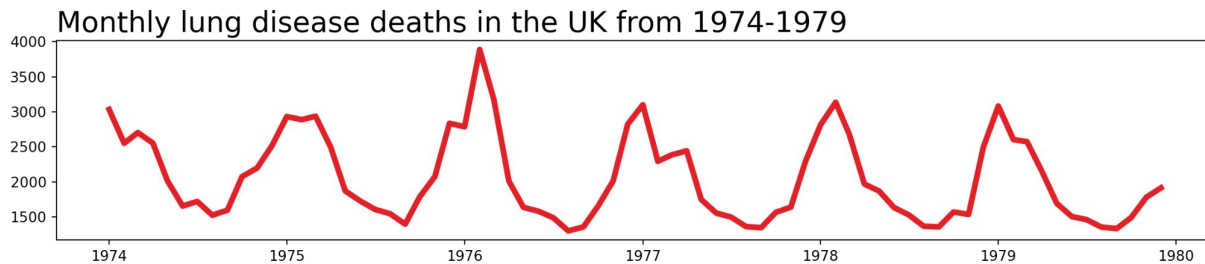
Partial Autocorrelation

x_t has a 7-day autocorrelation, but it also has a 14-day autocorrelation, 21-day autocorrelation, and so on.

x_t will have a 7-day partial autocorrelation but it will **not** have a 14-day partial autocorrelation since there is no overall 14-day trend that does not encompass two 7-day trends.

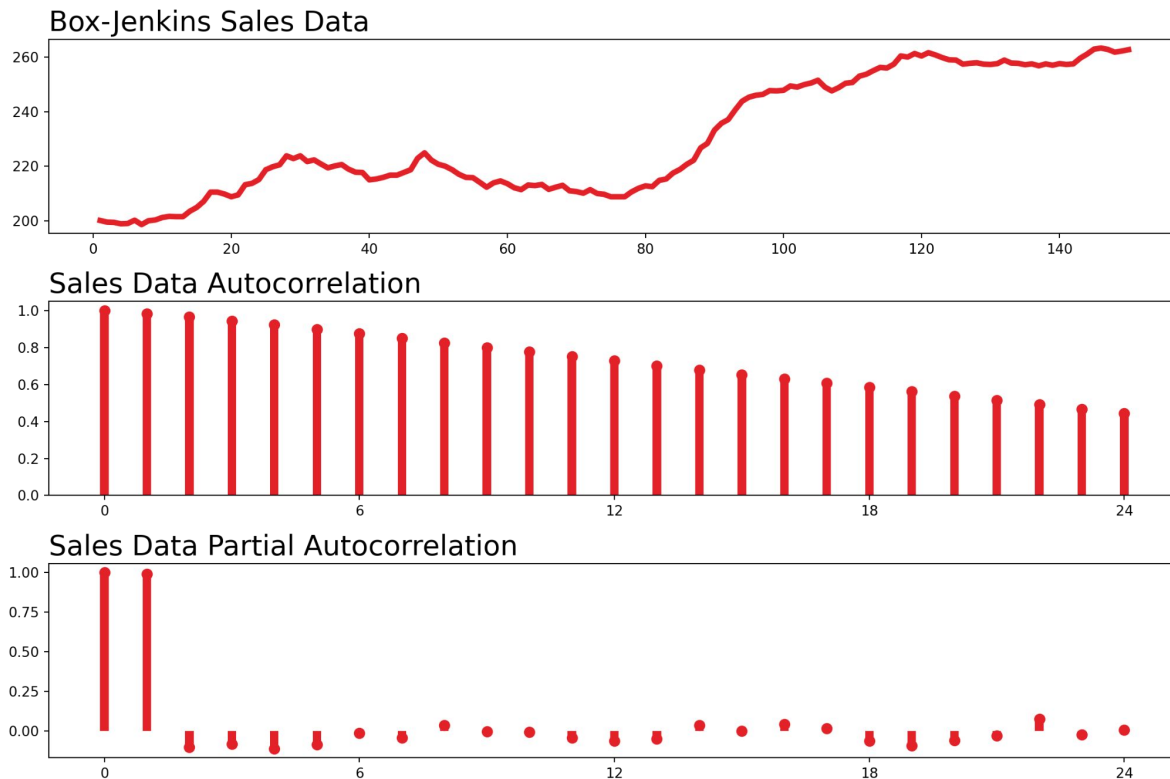


Example PACF: Lung Disease (monthly data)



(Notice it's
slightly flatter)

Example PACF: Sales (increasing trend)



Let's explore autocorrelations!

